



Review

Sensor-based safety management

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A B S T R A C T

The construction industry has one of the most hazardous working environments worldwide, which accounts for about 1 in every 5 occupational fatalities. The high rates of workplace injuries, illnesses and fatalities cause irreversible harm to workers and are often the source of delays and additional project costs. Improvements in sensor technologies, wireless communication, the processing power of computers, and advancements in machine learning and computer vision are now enabling the development of sensor-based safety management systems. The rapid growth of Building Information Modelling (BIM) has also created opportunities for improving safety management. While considerable progress has been made to improve construction safety, few studies have focused on the integration of sensor-based systems and BIM. This research, which is motivated by the development of such integrated methods, carries out a systematic review of the relevant literature, summarising recent developments of sensor-based safety management systems and advancements in safety management through BIM. The research gaps are identified and an outline for potential future research is provided. The results of the review reveal the potential of combining sensor-driven systems with BIM for improving safety management in construction.

1. Introduction

Around 15% of occupational fatal accidents in Australia have occurred on construction sites, where construction only accounts for 5% of the workforce [1]. In 2017, construction fatalities in the United States alone amounted to 971, which is more than 20% of all occupational deaths [2]. In the past few decades, research on safety climate and culture [3], worker-oriented safety [4], safety management programs [5], hazard recognition and risk assessment [6], and applications of information technology in construction safety led to improvements in overall safety performance [7]. However, the number of fatal and non-fatal accident injuries remains significantly high in the construction industry. Previous investigations have identified the lack of proactive and preventative measures such as hazard identification, control, and safety training as precursors to many construction accidents [8]. Another important factor, influencing the effectiveness of safety controls, is capturing and communicating details of the job site, which determines how efficient the knowledge can be absorbed [9].

Recent advancements in machine learning, computer vision, internet of things, and the increased affordability and processing power of advanced technologies have prompted researchers to work towards the development of sensor-based safety management systems, thereby

adding a technology-driven safety layer to construction sites. Given that safety is an ongoing issue on construction sites, the development of continuous safety monitoring systems has great potential to improve safety risk management. As described by Australia's Work Health and Safety Act (WHS), the risk management process involves the identification of hazards, assessing risks, controlling risks, and reviewing control measures [10]. Sensor-based safety management systems can improve each of these aspects of the safety risk management process as will be discussed in this review. On the other hand, Building Information Modelling (BIM) has been recognised as an effective tool for organising knowledge and enhancing communication [11]. There is potential to combine the two technologies for achieving better safety performance.

This research, which is motivated by the development of the aforementioned technologies, carries out a systematic review of the relevant literature with the following objectives: (1) summarising advancements in the development of sensor-based systems and their influence on the process of safety risk management; and (2) summarising the advancements in BIM and their influence on the process of safety risk management; and (3) identifying the research gaps in the literature, and outlining the potential future research directions. Section 2 describes the research methods employed in this paper. Section 3 provides

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an overview of the current literature on sensor-based safety management systems in construction. Various sensor-based safety systems are reviewed in detail, and their influence on each step of the safety risk management is discussed in Section 4.1. The integration of safety management with BIM is discussed in Section 4.2. Section 5 outlines the identified research gaps and opportunities for future research. The conclusion is presented in Section 6.

2. Research methods

This section provides the outline of the methods adopted to conduct the systematic review, exploring the most relevant studies surrounding sensor-based safety management systems and Building Information Modelling (BIM). The systematic review is based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [12], with a slight modification to adapt the process for the purpose of this study. Scopus database by Elsevier was selected for literature exploration, and three search attempts were taken to ensure that the identified publications thoroughly cover the focus of this study.

First, to identify the studies that have employed sensor-based systems in construction to improve safety, keywords such as *safety*, *sensor*, *camera*, and *construction* were used, and as a result, 328 documents were found. In the second round, keywords such as *construction management*, *building information modelling* and *safety*, and their equivalents were used, and Scopus returned 300 documents as of March 2019. In addition to the two search attempts described above, an explorative search was conducted to ensure the inclusion of the most relevant studies that would have otherwise been omitted due to search limitations. Therefore, the third search attempt included tracking citations of the most relevant articles and following their references. It further includes an explorative database search using keywords suggestive of sensor-based safety management systems involved in the studies, such as *construction safety management*, *machine learning*, *computer vision*, *sensors*, *information and communication technology*, *building information modelling*, etc. As a result, 243 articles were identified in the third

search attempt.

Fig. 1 illustrates the flowchart of the systematic review process based on the PRISMA flow diagram. The number of documents returned after applying each selection criteria is shown after each corresponding step.

As Fig. 1 illustrates, the results of the three search streams were then merged, and the duplicates were removed, amounting to a total of 798 documents. Next, the articles were screened based on their title. The most relevant studies were selected based on their consistency with sensor-based safety management in the construction industry. Therefore, the papers that contained the keywords used in the search process but were not relevant to the construction industry were removed. Further, as this study is mainly focused on safety management during the construction phase, studies that were aimed at facility management, structural safety or structural health monitoring were also excluded. As a result, the relevant studies were identified, leading to a total of 397 documents. The dataset containing 397 articles was used for the quantitative analysis, for which the results are discussed in Section 3.

Furthermore, the most relevant papers, selected from the dataset of 397 documents were identified based on their abstract and reviewed in a qualitative analysis, as presented in Section 4 of this study. Section 4 also contains a comparative review of the most influential articles from the mentioned dataset, as shown in Table 3. The selection criteria for the documents to be included in the comparative review are outlined as follows. For each year, only highly cited papers were included, with the exception of articles published in 2019. As the elapsed time since their publication, at the time of this study, was not sufficient to indicate their impact, all papers from the year 2019 were included. Using the above methods, a comprehensive analysis of the literature was conducted, resulting in the identification of gaps and opportunities for future research.

3. Analysis of publications

The database of the selected papers covers articles published from

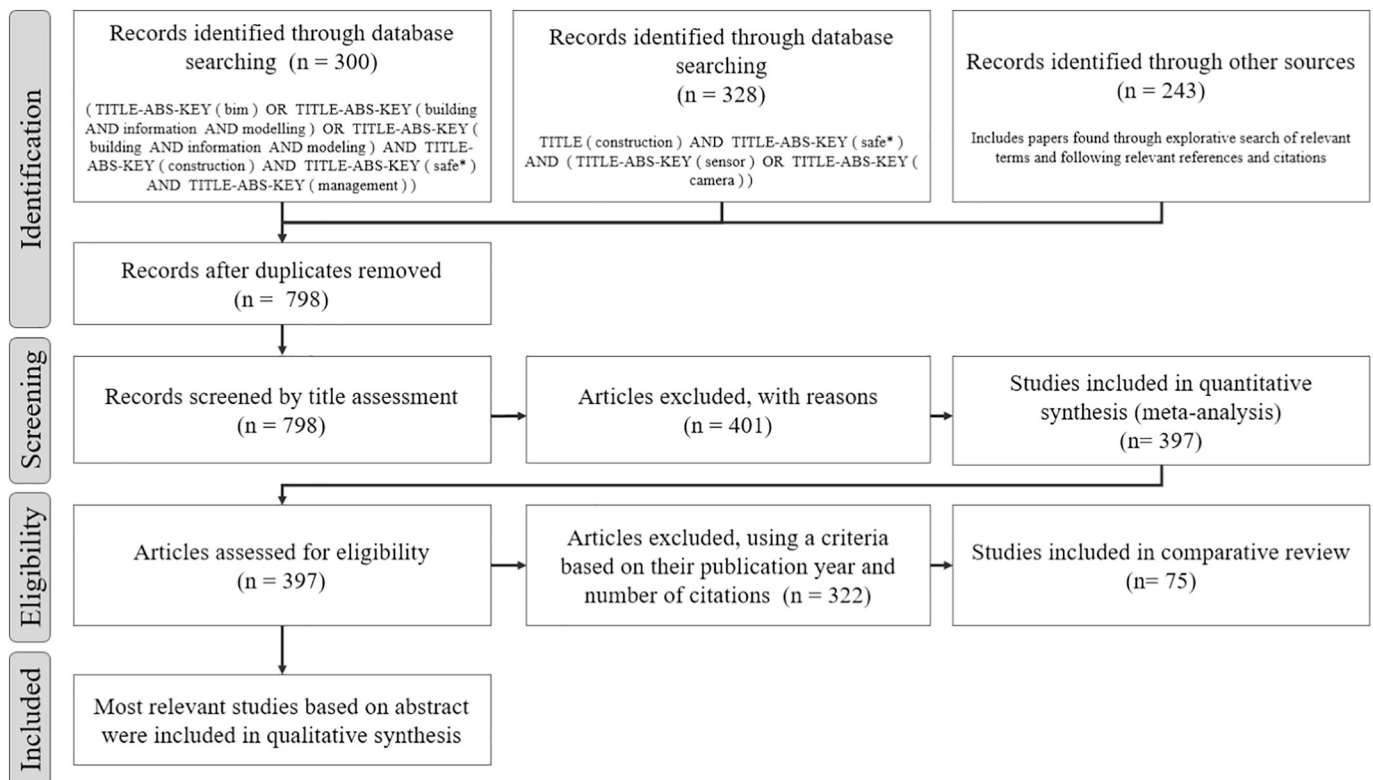


Fig. 1. The process of the systematic review (based on the PRISMA method).

Table 1
Previous review studies.

Authors (year)	Source title	Review method	Main focus	No. of reviewed studies
Zhou et al. (2013) [14] Seo et al. (2015) [15]	Construction Management and Economics Advanced Engineering Informatics	Systematic review State-of-the-art review	To review the technologies used in construction to improve safety management To assess various computer vision techniques used for safety monitoring in the context of construction	119 Not Specified
Zhang et al. (2017) [16] Zou et al. (2017) [17] Awolusi et al. (2018) [18]	Sensors (Switzerland) Safety Science Automation in Construction	Systematic review Critical review State-of-the-art review	To assess various sensor technologies used to improve construction safety management To review the studies surrounding BIM assisted risk management To examine the applications of wearable technologies for safety monitoring of construction workers	93 Not Specified Not Specified
Martínez-Aires et al. (2018) [19] Edirisinghe (2018) [20]	Safety Science Engineering, Construction and Architectural Management	Systematic review Systematic and critical review	To review the literature surrounding BIM and construction safety To evaluate various applications of sensor technologies in the broad context of construction	76 114
Antwi-Afari et al. (2019) [21]	Engineering, Construction and Architectural Management	Systematic review	To examine applications of sensing and warning based technologies for safety improvement	87

2003 to 2019. The number of relevant papers did not exceed 7 before the year 2011 and peaked at 99 papers in 2018. This is due to technological advancements and increased interest in using sensors, vision-based systems, and Building Information Modelling (BIM) to improve safety management in construction. Following this upsurge of interest, a number of review studies have been conducted. Table 1 categorises the most relevant reviews in three different groups, namely systematic, state-of-the-art, and critical review, as defined by Grant et al. [13], and outlines their main focus area. None of these studies, however, provide an overview of both the latest developments in sensor-based safety management systems and their integration with BIM. The objective of this study is to investigate the advancements in both of the mentioned research areas with an in-depth analysis of their influence on the four stages of safety management, namely hazard identification, risk assessment, control risks and review control measures and further, to identify research gaps and future trends, aimed to accelerate development of effective solutions.

The selected studies in this review include publications from more than 35 different journals. Table 2 lists the journals that have contributed three or more papers to the database constructed (397 papers). Automation in Construction has contributed the most to this domain, with a contribution of just under 100 papers, which is roughly 25% of the documents. Other highly relevant journals include Journal of Computing in Civil Engineering with 24 articles, Advanced Engineering Informatics with 21, Journal of Construction Engineering and Management with 19, and Safety Science with 18 publications. The research impact of the top 18 journals as of 2017 can also be viewed in Table 2.

Based on the database of selected articles, an analysis was performed to identify the most occurring keywords and the journal in which they appeared. Gephi, an open source graph and network analysis software [22], is used for this analysis. The resulting network is illustrated in Fig. 2, revealing the high occurrence of a variety of innovative techniques and technologies including computer vision, accelerometers, location tracking, RFID, and wireless sensor networks. The following section explores the undertaken approaches to improve safety management using these techniques in further detail.

4. Discussion

4.1. Sensor-based safety management

This section provides an overview of latest developments in sensor-based safety management systems and discusses their relation to various steps of safety risk management as described by Australia's Work Health and Safety Act (WHS) [10]. The innovative techniques and technologies employed in the development of sensor-based safety management systems in the last decade are presented in Fig. 3, illustrating the frequency of these techniques and technologies in the literature, distributed by year. The database of 397 papers, as described in Section 2 was used for this analysis. The figure shows a steady increase in the appearance of BIM in the literature. Computer vision techniques have also been used frequently in the literature, with their appearance gradually increasing from the year 2008, and showing a significant rise in 2018. Further analysis revealed a correlation between this sharp increase in the use of computer vision techniques, and the emergence of Convolutional Neural Networks (CNNs), which have highly improved the performance and robustness of the techniques [15]. Moreover, Fig. 3 demonstrates a rise in the deployment of all the relevant techniques and technologies from the year 2016 onwards, indicating a move towards sensor-based safety management in the construction industry. The following subsections cover the impact of such systems on identifying hazards, assessing and controlling risks, reviewing control measures in further detail and discussing various approaches.

4.1.1. Hazard identification (HI)

Hazard identification is the process of finding entities or situations

Table 2
Top Journals in sensor-based construction safety.

No.	Journal	No. of papers	CiteScore 2017	SJR Impact Factor 2017
1	Automation in Construction	99	5.36	1.613
2	Journal of Computing in Civil Engineering	24	2.07	0.697
3	Advanced Engineering Informatics	21	4.15	1.167
4	Journal of Construction Engineering and Management	19	2.36	1.023
5	Safety Science	18	3.22	1.113
6	Sensors (Switzerland)	10	3.23	0.584
7	Visualization in Engineering	6	N/A	N/A
8	Journal of Management in Engineering - ASCE	5	2.79	1.233
9	Procedia Engineering	5	0.89	0.282
10	Engineering Construction and Architectural Management	4	1.9	0.653
11	Journal of Civil Engineering and Management	4	1.86	0.572
12	Journal of Information Technology in Construction	4	1.73	0.589
13	Applied Ergonomics	3	2.95	1.071
14	Computer-Aided Civil and Infrastructure Engineering	3	4.97	1.154
15	Construction Innovation	3	2.12	0.731
16	Construction Management and Economics	3	1.66	0.816
17	Electronic Journal of Information Technology in Construction	3	1.73	0.589
18	KSCE Journal of Civil Engineering	3	1	0.329

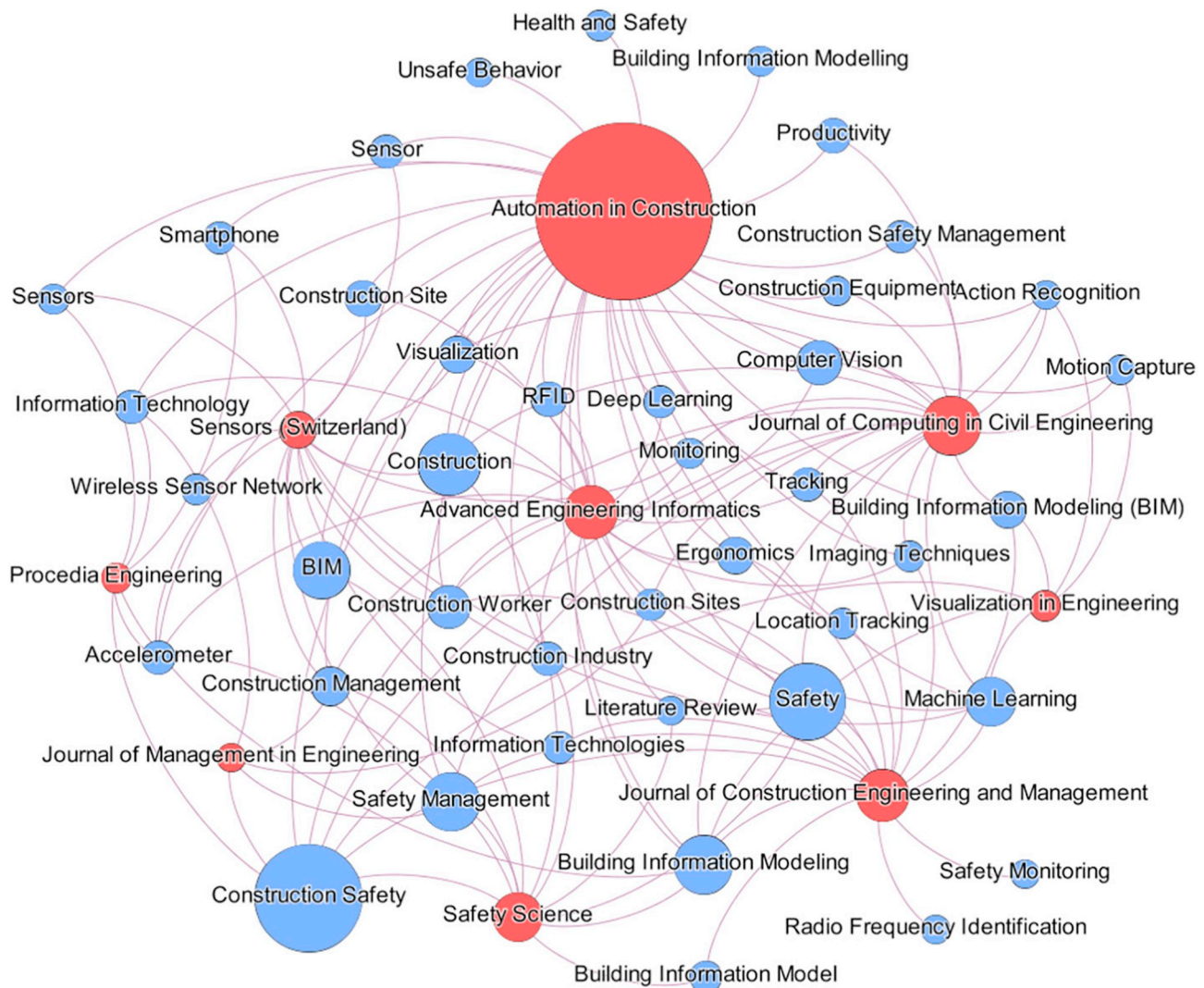


Fig. 2. Network of most frequently used keywords in relevant journals.

that have the potential of causing harm. Developing sensor-based systems for real-time hazard identification is a crucial step towards intelligent safety management. Three different approaches to hazard identification are discussed in this section. First, an indirect approach,

which attempts to identify hazards by analysing data from the construction site for anomalies that may be suggestive of an existing hazard. Second, identifying hazards as they directly relate to the behaviour of workers. The last approach is to monitor the workplace for

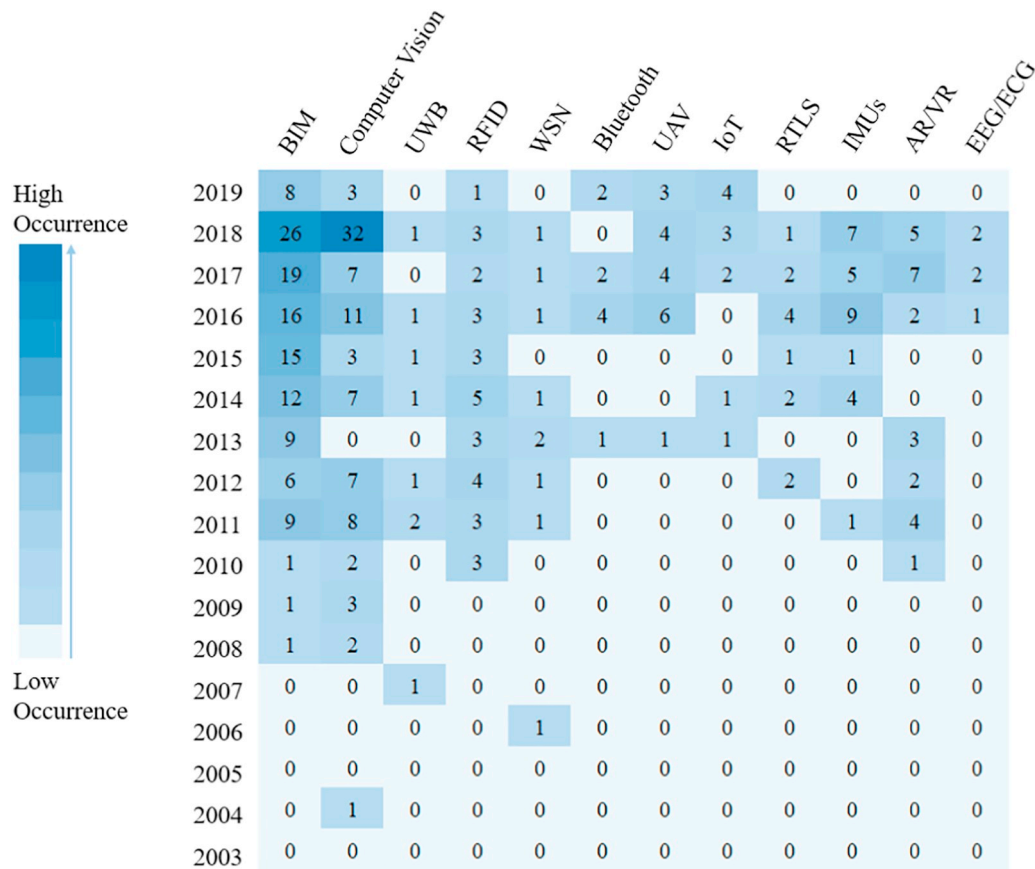


Fig. 3. Frequency of technologies used in the development of safety management systems in the literature (per year).

specific anticipated hazards.

4.1.1.1. Indirect hazard identification. Considering the dynamic nature of a construction site, regular safety inspections need to be carried out to ensure a hazard-free work environment. However, since inspections are often manually performed, it is not uncommon for new hazards to remain unidentified for a prolonged period of time, thereby exposing workers to risks of tripping, falling, being struck-by falling objects, electrocution or other accidents. Continuous monitoring of the site can thereby immensely facilitate safety management. An innovative approach to automatically identify hazardous areas has been proposed by Kim et al. [23]. In this study, a real-time tracking system is used to identify potential hazardous areas by comparing the actual path of workers to their optimal routes. Location tracking is performed using an RFID-based Real-Time Location Tracking System (RTLS), where RFID tags are mounted on workers' hardhats. Upon testing the framework in a case study, it was found that 80% of the identified hazards by the system corresponded to real hazards such as material piles, openings, areas with inadequate electric wire protection, and areas with a lack of falling object protection.

Although RFID technology is used in various industries for tracking some entities, the rapidly changing nature of the construction sites and their complexity make effective use of RFID very challenging. RFID requires a direct line of sight for optimum accuracy, which is a condition that is hard to realise in construction. In addition, it requires the installation of multiple tags and receivers which limit its practicality [24]. Ultra-wideband (UWB) is another radio frequency-based technology used for location tracking. UWB uses high-bandwidth radio pulses for communication, which makes it less susceptible to signal interferences. However, it still requires the installation and maintenance of multiple tags and receivers [18]. Global Positioning System (GPS) has also been widely used to track worker trajectories in

construction sites to improve safety. For instance, Arslan et al. [25] demonstrated a prototype of a GPS-based system for identifying stay points and the intersection of multiple trajectories, and classifying the trajectory into running or walking. Such semantic enrichment of data using various data mining and machine learning techniques helps to improve the decision-making process by providing insight on the behaviour of workers.

Vision-based systems as an alternative to other methods for tracking various entities on construction sites, has recently attracted the attention of researchers. Although still in the early stages of development, vision-based systems have the advantage of not requiring the installation of multiple tags, incur lower costs and are easier to maintain. Park et al. [26] propose a vision-based method that uses videos obtained from ordinary 2D cameras, and employs detection and tracking algorithms in a hybrid system for tracking entities. However, the method is limited to tracking workers across video frames and not capable of producing trajectories with respect to the workplace. In another study, Konstantinou et al. [27] developed a framework for obtaining 3D trajectories using multiple camera views. The proposed method uses three sequentially activated matching techniques. The output of a 2D tracking method is first checked using a motion matching technique, followed by geometry matching, and finally a template matching technique. In any step, if a match is found, further checking is terminated, thereby significantly reducing the required computational power. In a comparative study, Xiao et al. [28] evaluated the performance of a number of 2D vision-based tracking methods employed in the context of construction. Their study showed that although vision-based systems are promising, many challenges such as occlusion, clutter, illumination, and scale variation need to be tackled for the practical deployment of such systems.

Another innovative approach for identifying potential hazards based on real-time information obtained from the construction site was

proposed by Yang et al. [29]. They propose using wearable Inertial Measurement Unit sensors to analyse workers' gait abnormalities to identify potential falling hazards. In their experiment, subjects were asked to walk across a steel beam in a controlled environment. There were two hazards placed along the beam, namely a slippery surface, and an obstacle. Upon the analysis of data obtained from multiple subjects, it was found that the correlation between collective gait abnormality and the existence of a hazard is significant. However, this study was limited to one dimension only. The extension of such method to two or three dimensions by taking advantage of location tracking methods as discussed above would provide valuable information that could be used for effective real-time hazard identification. Moreover, such spatiotemporal data enriched with information about gait pattern, activity and so on can be used for further analysis.

4.1.1.2. Direct hazard identification (behaviour-related). Construction accidents are often directly related to the worker's behaviour. In an investigation for identifying the root causes of accidents, Abdelhamid et al. [30] explained that accidents occur either due to the failure of workers to identify hazards, and their decision to proceed despite the identification of hazardous condition, or their decision to act unsafely regardless of work conditions.

Real-time safety monitoring systems have great potential to be used as supplementary tools for safety supervision. Workers' disregard of hazards often manifests itself in the lack of adequate personal protective equipment usage. Recent advancements in computer vision and machine learning provide powerful tools for object detection using images. For instance, Park et al. [31] used the Histogram of Oriented Gradients along with Support Vector Machines to detect human bodies and hardhats, and by matching the geometric and spatial relationship between them, as well as identifying cases where the person is not using a hardhat.

Fang et al. [32] developed a method for monitoring the appropriate usage of a harness, anchoring, and hardhats by steeplejacks. In the proposed method, cameras are installed inside the rooms and facing the windows where steeplejacks are to perform the aerial work on exterior walls. They used a technique based on Convolutional Neural Networks (CNNs), known as Single Shot MultiBox Detector (SSD) for detecting improper usage of fall arrest systems. In a similar study, Fang et al. [33] used Faster R-CNN, and a deep CNN model to identify workers not wearing a safety harness when working at height. Using automated safety inspections, safety managers can be notified of unsafe behaviours, and take appropriate action to mitigate the risks in a timely manner.

A different approach of identifying unsafe behaviours was proposed by Guo et al. [34], who used depth cameras to identify various postures related to unsafe behaviour. Various body angles obtained from the skeleton-based posture information are compared to an unsafe behaviour database in order to identify postures that are suggestive of unsafe actions, such as jumping over a guardrail or dumping construction waste from higher levels.

Other forms of unsafe behaviour leading to hazardous conditions include working under the influence of drugs or alcohol, working under a high level of fatigue or attempting a hazardous task with a low relevant skill level. One example is the loss of balance possibly induced by the aforementioned factors. To address this issue, Umer et al. [35] developed a balance monitoring tool that uses wearable Inertia Measurement Units (IMUs) and fuzzy set theory to determine workers' balance performance by taking 20-second tests at different times of the day.

Monitoring the physiological condition of workers using Physiological Status Monitors (PSMs) has also been explored [36]. Using these methods, metrics such as heart rate and breathing rate can be monitored, providing valuable information for identifying unsafe working conditions as they arise. High levels of stress induced by schedule pressure or other factors may cause workers to act unsafely in

certain conditions. Therefore, it is essential to provide a work environment that is free of stress-producing conditions. Hwang et al. [37] and Jebelli et al. [38] explored the possibility of using electroencephalogram (EEG) to monitor the stress levels of workers. They validated their results by comparing them with saliva tests that examine cortisol levels, which is an indicator of emotional state.

4.1.1.3. Direct hazard identification (workplace-related). Certain hazards can be identified by monitoring the workplace for specific accident precursors. One example is site congestion. Congested working areas are more prone to accidents. Location tracking systems as discussed in Section 4.1.1.1 can enable workplace analysis in terms of congestion. Zhang et al. [39] used GPS sensors attached to workers' hardhats to track their activity while working on cast-in-place concrete columns. The workspaces were then visualised in a BIM platform to identify conflicts among other work-zones or material handling paths.

Environmental factors of the site such as temperature, noise level, and pollutants are significant workplace hazards that require constant monitoring. For instance, Riaz et al. [40] proposed using wireless sensor networks for monitoring workplace temperature conditions and integrating the system with a BIM platform for enhanced work and safety management.

Temporary structures on construction sites are another aspect requiring a thorough inspection to ensure all safety requirements are met. Safety inspection of scaffolds, for instance, is performed visually by the inspector and is a labour-intensive process. Cho et al. [41] proposed a system for real-time safety monitoring of the structure using strain sensors mounted on the scaffolds. With the aid of Finite Element Analysis (FEA), they developed a machine learning model to classify various states of the scaffolds such as over-loading, uneven settlement, and over-turning based on the data obtained from the strain sensors.

Vision-based systems also have the potential to be used as an effective monitoring method to ensure worksite safety. Kolar et al. [42] explored using transfer learning and CNN-based models for detecting guardrails in images. Furthermore, an improved version of Faster R-CNN can be used for detecting workers and equipment in images of civil infrastructure projects [43].

Often in construction sites, workers are required to perform their task alongside heavy equipment in a shared working area and are thereby exposed to high risk accidents. The development of proactive real-time alarm systems are another active area of research [44]. In a recent study, Soltani et al. [45] developed a method that combines computer vision based systems with real-time location systems to perform three dimensional excavator pose estimation, using surveillance cameras installed on the construction sites as stereo-cameras.

Understanding task-specific accident precursors, and defining quantitative metrics for the evaluation of unsafe behaviours or conditions will facilitate the development of effective real-time monitoring systems [15].

4.1.2. Risk assessment (RA)

Assessing risks associated with the identified hazards requires the consideration of how severe the consequences are if a worker is exposed to the hazard, and how likely it is to occur [46]. Sensor-based systems facilitate the process of risk assessment and reduce the bias caused by the subjectivity of the analysis.

The context in which hazards appear and the co-occurrence of various hazards determine the severity and likelihood of risks. For instance, in the case of struck-by accidents, heavy equipment approaching a worker in a congested working area is more likely to cause an accident than one approaching a worker in a spacious area. Similarly, the close proximity of a worker and equipment is more indicative of high risk if the worker is standing in the blind spot of the equipment [47]. Kim et al. [48] developed a method to monitor struck-by accidents using computer vision. They proposed using fuzzy inference for determining the safety level, based on risk factors such as congestion and

proximity.

Evaluating safety and health risks associated with a given task is another essential part of safety management. A number of studies have focused on methods of performing ergonomic analysis in a minimally intrusive way. The rapidly changing and complex nature of construction make reliable ergonomic assessment challenging. To address this issue, researchers have developed sensor-based systems to quantitatively perform ergonomic risk analysis [49]. The outcome of such systems can be used to improve safety training, planning, and task design.

For instance, insole pressure sensors have been used to detect and classify awkward working postures [50]. In a study by Nath et al. [51], they used built-in accelerometers and gyroscope sensors in smartphones, and machine learning algorithms to estimate the duration and frequency of various activities. Consequently, they evaluated the overexertion risk level associated with the task.

Vision-based systems have also been used to perform ergonomic risk assessment using posture analysis based on body angles. Golabchi et al. [52] proposed a framework for data collection, analysis, and visualisation to facilitate ergonomic analysis. Vision-based techniques that utilise CNNs, to estimate 3D skeleton of the subject from 2D images for ergonomic assessment have also been reported [53].

Risk assessment is a crucial step in safety risk management. Therefore, the development of effective automated safety monitoring systems requires the integration of sensor-based risk assessment systems into the frameworks.

4.1.3. Control risks (CR)

Controlling risks, once identified and assessed, can be categorised into three levels based on their effectiveness. The most effective level of control is elimination. Real-time safety monitoring systems facilitate the elimination control by identifying newly appearing or unidentified hazards and informing safety managers or site supervisors, which will then take the necessary steps to eliminate the risks if applicable.

The second level of control includes substitution of the hazard with a safer alternative, isolating the hazard from people, or reducing the risks through engineering controls. Safety will be enhanced by the timely identification of hazards, and notifying the people in charge to take appropriate actions. In addition, real-time safety alarm systems that will notify the workers or equipment operators of danger can be considered as another form of engineering control, driven by sensor-based monitoring systems.

The last level of control is aimed at reducing exposure to hazards. This level includes methods such as Personal Protective Equipment (PPE) and administrative controls. Proper usage of PPE is often overlooked on construction sites. Real-time monitoring systems provide a promising solution to the continuous supervision of their appropriate utilisation. The data obtained from real-time systems can also be used for trend analysis and evaluating the safety culture at different work zones or various stages of the project. This facilitates targeted administrative controls and enhances the quality of safety training [54].

4.1.4. Review control measures (RCM)

The last step in the safety risk management process refers to performing hazard identification, risk assessment and controlling risks repeatedly to ensure appropriate control measures are taken at all times. Sensor-based monitoring systems facilitate automation of the process. By continuously monitoring the workplace and workers' conditions, the automated safety risk management systems supplement human supervision to prevent accidents at construction sites effectively. The ongoing review of control measures is a crucial step in safety risk management. BIM, as illustrated in Fig. 4, provides a platform for improving various aspects of safety management, in addition to creating an opportunity for integrating sensor-based safety management systems. The next section will discuss the integration of safety management with BIM in further detail.

4.2. Integration of safety management with BIM

BIM has become pivotal to the Architecture, Engineering, and Construction (AEC) industry. It has increased collaboration between project stakeholders, thereby improving the process of design and construction [55]. In the past two decades, several additional dimensions have been added to BIM beyond the geometrical dimension, expanding its functionality. In a systematic review, Charef et al. [56] demonstrated that the 4th and 5th dimensions of BIM are unanimously referred to time and cost, respectively. However, they found that some industry practitioners and academics relate the 6th and 7th BIM dimensions to sustainability and facility management, and to a lesser extent safety, with no general consensus. It is apparent that currently, no commercial BIM platform with an embedded comprehensive safety management dimension is available [57,58].

Fig. 5 illustrates the network of most frequent keywords appearing in the constructed database of 397 articles, discussed in Section 2. The size of the nodes is directly proportional to their frequency of occurrence, and the edges represent their co-appearance with other keywords. It appears that "BIM" is the most occurring keyword, and its co-appearance with a variety of techniques and technologies indicate that several studies have been performed to examine the integration of BIM with sensor-based safety management systems. The following sections explore various approaches to integrating safety with BIM. The approaches are grouped into four categories: (1) BIM for knowledge management; (2) BIM for safety planning; (3) BIM assisted design for safety; and (4) BIM for real-time safety monitoring.

4.2.1. BIM for knowledge management

Knowledge management in any organisation is crucial for maintaining a competitive advantage, as it is an essential factor for achieving high work efficiency [59]. Construction projects are unique, fragmented and complex, requiring temporary and multidisciplinary teams. Therefore, effective knowledge transfer would prevent similar safety breaches, reduce risks of safety hazards, and enhance safe work practices [60]. In addition to safety knowledge in the form of personal experience, and project-specific information, numerous safety rules and regulations, best practices, and accident records exist in diverse and fragmented formats [61]. Moreover, as discussed in Section 4.1, emerging sensor-based safety management systems will create enormous datasets, which need to be processed, analysed and transformed into knowledge. Bilal et al. [62] identified several potential opportunities for utilising Big Data within BIM, such as linking BIM with external sources, using BIM for the Internet of Things (IoT) applications, and enabling Big Data storage systems that are compliant with Industry Foundation Classes (IFC). The research surrounding knowledge management can be categorised into three groups: (1) knowledge capture and reuse; (2) knowledge sharing; and (3) knowledge storage and retrieval [11]. The following paragraphs will discuss relevant studies addressing different aspects of knowledge management with the potential to improve safety.

BIM platforms are developed using an object-oriented structure to enable the utilisation of customised parameters in the models for capturing additional information related to a specific building element. In addition, BIM as the central platform increases reusability of the captured knowledge. Elghamrawy and Boukamp [63] proposed using OWL-based ontologies for indexing information and leveraging RFID tags installed on various building components for quick access to related knowledge. For instance, information about a particular solution that has been proven effective in previous projects can be easily retrieved when encountering a similar problem. Similarly, the information concerning safety as related to the construction of a building element can be stored as a parameter of the corresponding building element in the BIM model, thereby facilitating the capture and reuse of relevant safety knowledge. Another approach is to leverage the capabilities of external applications, and through the Application

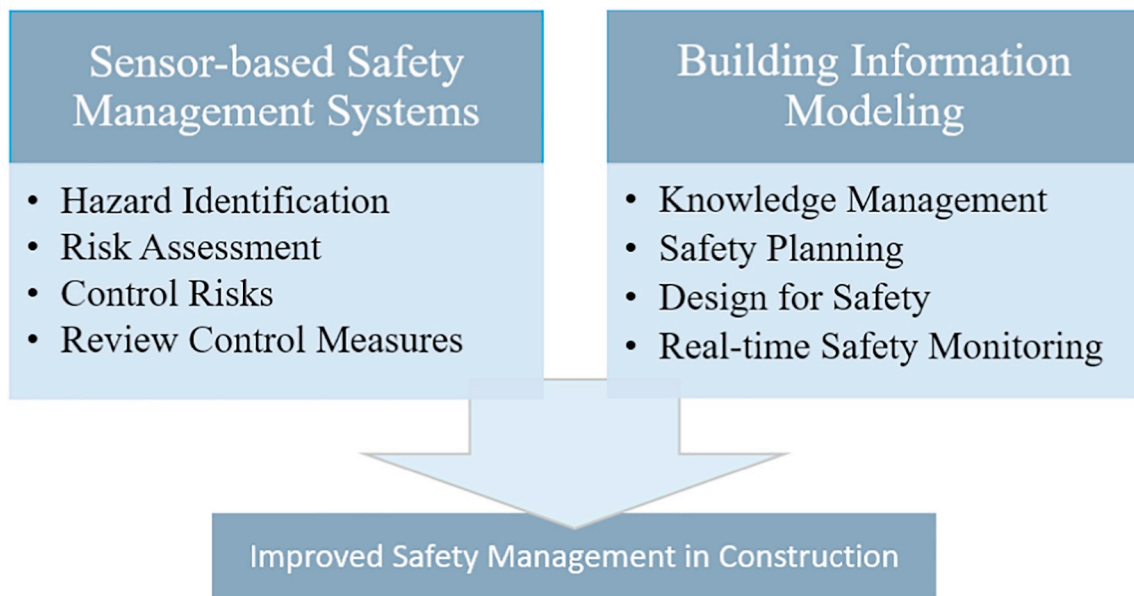


Fig. 4. Hybrid use of sensor-based safety management systems and BIM.

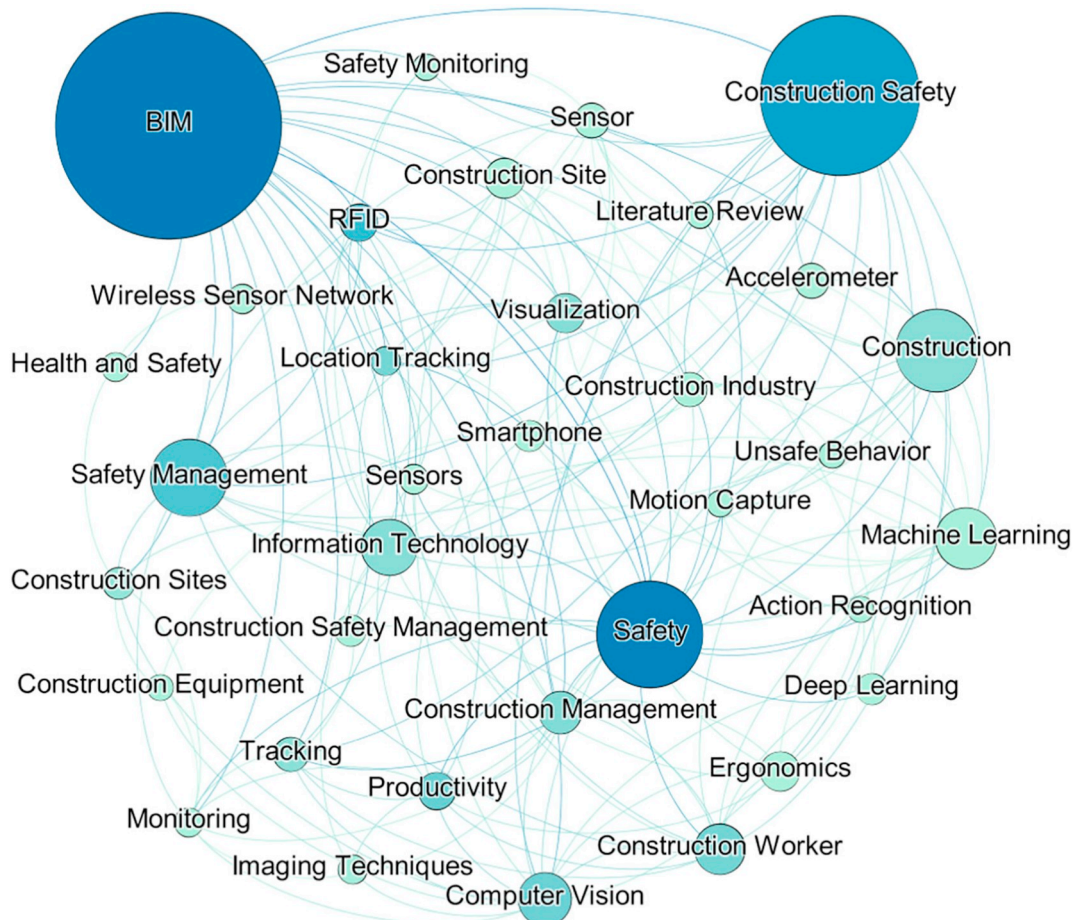


Fig. 5. Network of frequently used keywords in the 397 papers reviewed.

Programming Interface (API) of BIM software, integrate the solution with the BIM model [61]. For example, Ding et al. [64] used ontologies and semantic web technology to organise the relevant risk knowledge semantically and through the BIM platform, and created a link between various building elements and the proper knowledge, thereby

facilitating knowledge reuse in the risk analysis process.

As defined by Alavi and Leidner [65], knowledge sharing entails the dissemination and exchange of knowledge between groups or individuals. Construction projects are dynamic and fragmented by nature, and involve segregated teams working simultaneously. Various teams

enter the project at different stages of construction to perform a specific task. Therefore, effective communication is crucial for the optimal delivery of the project. Visualisation techniques can significantly improve the effectiveness of knowledge sharing. Benefits of using visualisation for communication among different teams, and between the site supervisor and workers include the enhancement of working memory [66,67], improved communication of spatial information [68], better utilisation of knowledge from past experience and memory [69], and reducing the negative effect of a language barrier on effective communication [70,71]. Along this line, several studies have explored leveraging the visualisation capabilities that BIM provides, and technologies such as augmented/virtual reality (AR/VR) for enhancing safety communication and safety training [72–74]. Taking a data-driven approach, Mo et al. [74] proposed a framework that utilises a clustering technique to analyse accident reports and used this information to develop safety training programs. Li et al. [75] proposed another innovative sensor-based framework, who leveraged a real-time location network system and BIM to improve safety training by providing feedback to workers performing a precast installation task.

As the amount of available data increases exponentially, storage and retrieval of knowledge from the large volumes of information now available also become a challenge. Although a number of Big Data techniques have been explored in the construction industry, its adoption remains relatively new in the industry [62,76]. For example, You and Wu [77] proposed a big data infrastructure for enhancing data interoperability between Enterprise Resource Planning (ERP) systems and BIM. The unified big data platform thereby facilitates knowledge discovery and decision making using historical data accumulated over time. As part of Big Data Engineering, selection of an appropriate Big Data storage solution plays a critical role in the development of sensor-based systems. To find the most suitable database solution for a BIM-integrated wireless sensor network, developed for monitoring safety in confined spaces, Riaz et al. [40] tested and compared different types of database models such as SQL, NoSQL and time series databases, and found MongoDB to outperform its alternatives for this particular application. On the other hand, retrieval of knowledge from large amounts of data has been made possible through Big Data Analytics, which includes techniques such as data mining, machine learning, regression, classification, clustering, and Natural Language Processing (NLP) [62]. A number of studies have explored the utilisation of such techniques for injury prediction, identifying attributes of a project that correlate with injuries, and discovering accident precursors by retrieving knowledge from injury reports [78–80]. Further research is required to increase the robustness and reliability of automatic knowledge retrieval techniques.

4.2.2. BIM for safety planning

Construction safety planning can be viewed as a two-step process, which consists of identifying hazards, and consequently making a decision on selecting the best possible safety measure before the commencement of the construction task [81,82]. Currently, safety planning is mostly performed manually by relying on 2D drawings, fragmented safety regulation documents and the experience of safety managers [61]. Moreover, the practice is not highly responsive to schedule changes, and often lags behind or fails to adapt to dynamic schedule changes [83]. BIM as discussed in Section 4.2.1, in addition to storing the geometric model, project schedule, and material specifications, can also be used to facilitate the management of safety-related knowledge. Therefore, BIM has the potential to be used as a centralised platform to conduct safety planning efficiently.

A number of studies have utilised rule-checking approaches with BIM to automate the hazard identification and safety planning process for temporary structures. For instance, Zhang et al. [84] proposed using BIM with the format of IFC for increased interoperability, the detection of fall hazards, such as unprotected openings and holes, and identifying the appropriate control based on the regulations provided by

Occupational Safety and Health Administration (OSHA). The process requires the interpretation of rules into a machine interpretable IF-THEN format and preparation of the building model for executing these rules. Finally, the safety manager can use the automatically generated report to apply the necessary corrections. In another study, Zhang et al. [85] further demonstrated the effectiveness of the framework by implementing it in two case studies and comparing the process with the manual approach. However, the system requires the BIM model and schedule to be up-to-date, accurate, and complete. Similarly, Kim and Teizer [86] automated the scaffold design and planning process by implementing a rule-checking technique. Therefore, the required scaffolding materials can be estimated to higher accuracy, and the visualised scaffolding systems in the BIM model facilitate communication between project stakeholders, thereby reducing the risk of planning errors and unexpected hazardous events. BIM-based rule checking systems have also been used to automatically plan for the optimal placement of temporary stair towers [87]. However, reliable performance of such methods highly depends on the complexity of the models, and more robust algorithms need to be developed to render such systems practical.

Construction site layout planning is another aspect of safety planning that highly benefits from the capabilities of BIM, sensor-based safety management systems, and integration of the two technologies. In addition to improving the efficiency of the construction work, optimising personnel and material travel distance, and reducing cost [88,89], the process of site layout planning eliminates many potential hazards. Schwabe et al. [90] demonstrated a framework to automatically check the construction site layout to ensure compliance with the safety rules using the IFC model and an open source rule engine. However, this approach is contingent on model requirements being strictly followed. Another crucial aspect, which is often neglected in the decision-making process of site layout planning, is taking various risk factors such as hazards of falling objects, hazardous chemicals, and the risk of air and noise pollution into consideration. To address this, Ning et al. [91] proposed a quantitative risk assessment model to assist site managers in making better decisions in the face of various site layout scenarios. The model views the risk factors to be a product of either interaction flows, such as material, personnel or equipment flows, or arise from safety or environmental concerns, such as site waste, hazardous materials and equipment. The proposed model, however, considers site layout as a static problem and is only applicable to the preconstruction stage.

Several frameworks for utilising 4D BIM to enhance work-space planning have also been proposed [92]. The framework enables the identification of potential site congestions by using the characteristics of the construction plan, work-space, and activities. A number of other studies have also developed methods, leveraging site-specific spatial and temporal information handling capabilities of 4D BIM software in an attempt to improve safety planning processes [93,94]. Choe et al. [94] proposed a framework for estimating the risk level of various construction zones on different days, based on the project schedule, geometric model and general safety knowledge. The safety score is estimated based on the number of workers active in each zone and their occupation, where the relative safety score of each occupation is based on an earlier study conducted by Choe et al. [95].

In reality, when performing safety planning at different stages of the project lifecycle, the dynamic nature of construction cannot be overlooked [96,97]. As discussed in Section 4.1, the advancements in information technology and sensor-based systems create an opportunity for the real-time assessment of risk factors, and context-aware systems. For instance, Zhang et al. [98] used location data obtained from GPS data loggers attached to workers' hardhats to determine and visualise the occupancy position of workers performing a formwork removal activity around a concrete column. This demonstrates the feasibility of using real-time data from the construction site for performing safety planning more effectively. Golovina et al. [99] also experimented with

spatiotemporal GPS data, and focused on struck-by accidents in an attempt to determine the root causes of such accidents, by monitoring the proximity of workers to equipment, and when they enter the equipment's blind spot areas. Using this information, and by creating a heat map that visualises the areas where the workers' views are obstructed, safety planning can be performed more effectively. Further research is required to explore the integration of real-time sensor-based systems, BIM and their implications on safety planning.

4.2.3. BIM-assisted design for safety

As discussed in Section 4.1.3, the most effective safety risk control method is elimination. A significant number of safety risks that may arise during construction or at later stages of the project lifecycle are inherent to the design and can be eliminated [100]. These conditions that may lead to construction accidents are often identifiable during the design phase [101]. This has led several countries to pass legislations, encouraging and/or requiring designers to consider safety implications of the design on all stages of the project lifecycle, including the construction phase [102]. Australia (the Work Health and Safety Acts and Regulations), Singapore (the Workplace Safety and Health (Design for Safety) Regulations 2015), the UK (the Construction Design and Management Regulations 2015) and EU countries (adaptations of European Framework Directive 92/57/EEC) are among the countries that have introduced such legislations. However, it is worth noting that the US has no legislation regarding design for safety in place [102]. Identification of design choices with inherent safety problems has been significantly facilitated by the use of BIM, as it enables analysis of all the building elements and their relations in a single platform [103]. A number of studies have focused on the development of BIM-based tools to identify violations of safety standards using rule-checking methods and to provide decision support in the early stages of construction [102]. The process requires safety rules to be translated into a machine-interpretable language, and correlations between various building elements and safety rules need to be specified [104].

For instance, Cooke et al. [105] developed a prototype for assisting design engineers in making safety-conscious decisions, utilising argument trees to estimate the likelihood of a hazard based on certain design choices. Mirahadi et al. [106] developed a framework for evaluating the evacuation performance of a building using two risk indices, crowd simulation, and BIM in the format of IFC to assist designers in optimising the building layout for better safety performance in case of an emergency. Yuan et al. [107] created a Prevention through Design (PtD) knowledge base using safety design regulations, documents, and best practices. Using Microsoft Access as the database management system, and a plug-in developed for Revit, they created an interactive interface, which enables querying of the PtD knowledge base through the BIM platform.

The majority of the developed design for safety tools rely on explicit knowledge, which is often stored in the form of regulations, documents, and best practices. However, tacit knowledge (e.g. experience-based knowledge of construction workers), which is accumulated during each project is often neglected, and not communicated with designers [103]. Therefore, to address this gap, frameworks for facilitating the capture of tacit knowledge need to be developed. In addition, further research is needed to explore the possibility of using data analysis and data mining techniques to identify the correlations between various design choices and construction accidents. The emerging sensor-based safety management systems could provide the large amounts of data necessary for the implementation of such methods, leading to the identification and elimination of preventable safety hazards rooted in the design phase before construction begins.

4.2.4. BIM for real-time safety monitoring

The emergence of new innovative technologies and techniques, as discussed in Section 4.1 is transforming the process of safety management by augmenting it with sensor-based systems [20]. Many of the

developed frameworks, however, are standalone applications. In recent years, BIM has significantly enhanced the process of project delivery [108,109]. BIM provides geometrical information at the component level, as well as enabling effective management of metadata on various aspects of the project. In addition to spatial information, it allows for the incorporation of the project schedule, which provides numerous benefits to project management. Therefore, BIM is a highly suitable platform for the integration of sensor-based safety management systems. The information obtained from various sensor-based monitoring systems can be incorporated with BIM for real-time monitoring of the construction site. The visualisation capability of BIM renders it as an effective tool for the efficient communication of real-time information with site managers.

In addition to automated safety monitoring capabilities that this approach provides, in case of an arising unsafe and hazardous situation, safety managers can be quickly notified of the exact location, and the current state of the site, and thereby act in a timely manner to address the hazard and take appropriate measures to control the risks. A number of studies have explored using wireless network systems in conjunction with BIM for enhanced real-time safety monitoring. In these studies, environmental factors such as temperature, humidity, and oxygen levels are monitored and visualised within a BIM platform [110–112].

Dong et al. [113] developed a framework for integrating real-time location systems with pressure sensors embedded in personal protective equipment (PPE) to identify and monitor the misuse of PPE. Using a BIM environment accessible through a web application, the managers can specify danger zones, and the system can automatically identify the misuse of PPE in hazardous areas. In another study, Park et al. [114] used Bluetooth Low Energy (BLE) mobile tracking sensors to track workers' locations in real-time. Hazardous areas are specified in the BIM platform by the managers, and the presence of workers in unsafe zones is automatically detected.

Sensor-based safety management systems, as discussed in detail in Section 4.1, are not limited to environmental monitoring, and location tracking. The developed frameworks in the literature range from worker physiological state monitoring, the detection of unsafe behaviours, breaching safety regulations, improper use of personal protective equipment, the proximity of workers to heavy equipment, poor housekeeping, gait instability of workers and so on. Table 3 illustrates different technologies used for these applications and comparatively demonstrates various aspects of safety risk management (i.e. Hazard Identification (HI), Risk assessment (RA), Control Risks (CR), and Review Control Measures (RCM)).

5. Research gaps and future directions

The latest developments in sensor-based systems and their relationship to procedural safety risk management were reviewed, and the possibility of integrating such systems with BIM was investigated. This section identifies several research gaps and suggests future research directions.

In recent years, correlation of certain measures of safety management with injury rates, known as safety leading indicators, have been studied to identify the best predictors of future performance [167]. In a recent study, Alruqi et al. conducted a review of studies on safety leading indicators [168]. To formalize the domain and facilitate future research, they offered a definition of active and passive safety leading indicators, distinguishing between the two in the way they are measured. While passive leading indicators do not typically change during construction and are either present or absent (e.g. training or orientation, owner involvement, incentives etc.), active leading indicators are measured regularly (e.g. monthly) and may change during the construction process [168]. For instance, frequency of safety inspections, hazard reporting and safety meetings are some examples of active safety leading indicators used as predictors of future performance.

Table 3
Comparative review of key areas addressed, and their feature techniques or technologies.

Source	HI	RA	CR	RCM	BIM-integrated	Featured technology/technique	Notes
[115]	X	.	x	.	.	Internet of Things (IoT), Zigbee, WiFi, LTE	Monitoring unauthorised entry to a hazard zone
[108]	.	.	.	.	x	IoT, BIM	Review paper
[116]	X	X	x	x	x	RTLS, UWB, BIM, IFC	Near-miss/proximity analysis
[90]	X	X	x	.	x	BIM, Business rule management system (BRMS)	BIM-integrated rule-based system for safe construction site layout planning
[117]	X	.	.	.	.	Computer vision, deep learning, CNNs	Detection of construction workers
[111]	X	X	x	x	x	BIM API, IoT, WSN, data analytics	Sensor-based system for monitoring environmental conditions in confined work spaces
[118]	X	X	.	.	.	Electrodermal activity (EDA)	Assessing workers' perceived risk through monitoring workers' physiological response
[106]	.	.	.	.	x	BIM, Industry Foundation Classes (IFC)	Design for safety, fire evacuation
[119]	.	.	.	.	x	BIM, Knowledge management	Uses BIM to store and present safety knowledge
[120]	X	X	x	.	.	Bluetooth Low Energy (BLE), IoT	Monitoring safe usage of harness and proximity to risk zones
[121]	X	.	x	x	.	IoT, RFID, Infrared sensors, Smartphone, Web application, Cloud server	Detecting breaches of hard-hat-use
[122]	X	X	.	.	.	Computer vision, deep learning, CNNs, UAVs	Proximity monitoring for struck-by hazard identification
[123]	X	X	.	.	.	Indoor Positioning System (IPS), Bluetooth	Semantic enrichment of worker trajectories on construction sites
[124]	X	.	.	.	.	Computer vision, deep learning, CNNs	Detecting breaches of hard-hat-use
[125]	X	.	.	.	.	Computer vision, deep learning, CNNs, LSTM	Detecting unsafe actions, ladder usage monitoring
[18]	X	.	.	.	.	Wearable sensors, IMUs, EEG, ECG, RFID, UWB, Infrared, GPS, Bluetooth	Review paper
[19]	X	.	.	.	x	BIM	Review paper, BIM and safety management
[126]	X	.	.	.	.	Computer vision	Review paper, abnormal behaviour detection
[127]	X	X	x	x	.	IoT, Wearable sensors	Proximity monitoring for struck-by hazard identification
[128]	.	X	.	.	.	IMUs	Gait stability analysis
[112]	X	X	x	x	x	IoT, WSN, BIM	Monitoring for hazardous gas detection
[129]	X	.	.	.	.	Computer vision, deep learning, CNNs, transfer learning	Construction equipment detection
[33]	X	.	.	.	.	Computer vision, deep learning, CNNs	Safety harness detection
[43]	X	.	.	.	.	Computer vision, deep learning, CNNs	Worker and equipment detection
[130]	.	.	.	.	x	Computer vision, BIM	Indoor progress monitoring
[131]	X	.	x	.	.	Computer vision, deep learning, CNNs	Detecting non-certified work
[132]	x	.	.	.	.	Computer vision, deep learning, CNNs	Activity recognition
[133]	x	X	x	x	.	Wearable sensors	Review paper
[134]	x	X	x	x	.	Bayesian Networks	Analysing the influence of environmental factors on workers' behaviour
[135]	.	.	.	.	x	Point clouds	Progress monitoring
[136]	.	.	.	.	.	Robotic Construction Worker (RCW)	Eliminating worker hazard exposure by remote operation
[114]	x	X	x	x	x	Bluetooth Low Energy (BLE), IoT, location tracking	Hazard zone monitoring, location tracking
[17]	x	X	x	x	x	BIM, Knowledge management	Review paper
[137]	x	X	x	x	x	Visualisation technology, location tracking	Review paper
[138]	.	.	.	.	.	IoT	Review paper
[139]	x	X	.	.	.	IMUs, motion capture	Awkward posture recognition, work-related musculoskeletal disorders (WMSDs), ergonomics
[49]	x	X	.	.	.	IMUs, smartphones	Work-related musculoskeletal disorders (WMSDs), ergonomics
[140]	.	.	.	.	x	RFID	Improving schedule performance, risk management
[29]	x	X	x	x	.	IMUs	Collective gait analysis for hazard identification
[141]	x	X	.	.	.	Electroencephalogram (EEG)	Fatigue monitoring, physiological monitoring
[142]	x	.	.	.	.	Computer vision, Depth camera	Detecting unsafe behaviours
[143]	x	X	.	.	.	Electroencephalogram (EEG)	Assessing construction workers' vigilance
[144]	.	.	.	.	x	Unmanned Aerial Vehicles (UAVs), Computer vision	Review paper
[62]	.	.	.	x	.	Big Data Analytics	Review paper
[145]	x	X	x	x	x	RFID, BIM	Indoor localisation
[146]	.	.	.	.	.	Computer vision	Anomaly detection
[147]	x	.	.	.	.	IMUs	Work-related musculoskeletal disorders (WMSDs), ergonomics
[148]	x	X	x	x	.	IMUs	Near-miss fall detection
[64]	.	.	.	x	x	BIM, ontologies, semantic web	Knowledge management
[149]	x	X	.	.	.	IMUs, gait stability	Fall risk estimation
[150]	x	X	x	x	.	Systematic review	Review paper on safety management in construction
[84]	x	.	.	.	x	BIM, 4D, rule-checking	Fall hazard identification, safety planning
[61]	x	X	x	.	x	BIM, ontologies, semantic web, rule-checking	Job hazard analysis
[15]	x	.	.	.	.	Computer vision	Review paper, vision-based safety monitoring
[151]	x	X	x	x	x	RTLS, simulation	Proactive behaviour-based safety management system
[152]	x	X	.	.	.	RFID, RTLS	Indoor localisation
[86]	.	.	.	.	x	BIM	Safety planning for scaffolding
[110]	x	X	x	x	x	IoT, WSN, BIM	Monitoring environmental condition in confined spaces
[153]	x	X	.	.	.	Computer vision	Crane pose estimation
[154]	x	.	.	.	x	BIM, 4D	Schedule and workspace conflict identification
[155]	x	X	x	x	.	Sensors, robotics, Information analysis	Review paper
[92]	x	X	.	.	x	BIM, 4D	Work space planning
[156]	x	X	x	x	x	BIM, simulation, visualisation	Pro-active safety analysis
[157]	x	X	x	x	.	GPS	Pro-active safety analysis
[158]	x	.	.	.	.	Computer vision, motion capture	Unsafe behaviour detection, ladder safety
[70]	x	X	.	.	x	BIM, AR, visualisation	Enhance safety communication between construction managers and workers

(continued on next page)

Table 3 (continued)

Source	HI	RA	CR	RCM	BIM-integrated	Featured technology/technique	Notes
[159]	.	.	.	.	.	RTLS, Physiological Status Monitors (PSMs)	Work sampling
[160]	x	X	x	x	x	BIM, 4D, visualisation	Pro-active safety analysis
[161]	x	.	.	.	.	Computer vision, motion capture	Unsafe behaviour detection, ladder safety
[162]	x	X	x	x	.	IoT, RFID	Location tracking
[163]	x	X	.	.	.	Computer vision	Ergonomics, awkward posture detection
[164]	x	X	x	x	x	Databases, VR, GIS, BIM, sensors	Review paper
[165]	x	.	.	.	.	Computer vision, Histogram of oriented gradients (HoG)	Worker detection
[24]	x	X	.	.	.	RTLS, RFID	Location tracking
[166]	x	.	.	.	.	Computer vision, Histogram of oriented gradients (HoG)	Equipment detection

Acronyms: LSTM, long short-term memory; AR, augmented reality; UWB, ultra-wideband; CNN, Convolutional Neural Network; WSN, wireless sensor network; LTE, long-term evolution; RTLS, real-time location system; ECG, electrocardiogram; API, Application Programming Interface; IMU, Inertial Measurement Unit.

However, there is a gap to include insights obtained by sensor-based safety management systems in such studies, investigating the correlation of data-driven safety performance indicators with future safety performance.

Sensor-based systems have the potential to capture safety risk exposure continuously, providing a more realistic measure of safety performance, which can be used as an active safety leading indicator. To bridge the gap between the proposed techniques and their adoption by practitioners, the solutions are required to be integrated with existing practices with minimum operational expertise and maintenance required. Future research in the domain can be greatly enhanced by considering integration of such systems with BIM, offering end-to-end solutions. Currently, most studies have focused on the development of monitoring systems that collect and process certain types of information from various entities at construction sites (e.g. workers' gait pattern, physiological state, workers' activity, workspace congestion, temperature and so on). A desirable feature would be a framework for combining such information in an integrated system. Further research is required to explore how such information can be turned into actionable insight most effectively. Future research may also consider design of data-driven safety decision support platforms for site managers. This would accelerate adoption of sensor-based safety management methods by the industry.

As construction industry experiences an increase in adoption of sensor-based systems, more real-world data may become available to construction companies. This opens up further research opportunities as real-time monitoring systems create enormous amounts of data. Further domain-specific studies, including big data engineering and data analytics, are required to extract relevant information from data obtained from real-time monitoring systems, and explore their applications to construction safety [62]. Furthermore, visualisation of information obtained via real-time monitoring systems through BIM platforms can improve analysis, facilitate accessibility, and enhance communication and training.

Near-miss events are important safety-leading indicators that are often left unreported and undocumented. Although frameworks that facilitate the process of reporting near-misses and visualising them through BIM have been proposed [169], the possibility of using real-time monitoring systems to identify and document near-misses automatically has not yet been fully explored. Exploration of near-miss incident reports and identification of detectable incidents can prove to be highly beneficial to the development of sensor-based safety management systems. This gap can be filled by conducting accident investigations with the aim of identifying detectable accident precursors as they relate to falling, struck-by, caught in/between accidents, and electrocution. Development of sensor-based safety management systems will be accelerated by creating a detailed breakdown of each accident precursor, and defining quantitative parameters to monitor.

While some studies have explored the integration of wireless sensor networks with BIM [110], solutions that combine vision-based monitoring systems with BIM platforms have not been sufficiently explored.

Rapid advancements in the field of computer vision and an abundance of cameras on construction sites create a unique opportunity for vision-based techniques to be used as effective monitoring systems in the industry. However, the majority of vision-based studies have tested their proposed algorithms on datasets that are proprietary to that project and are often small in size. This causes the solutions to overfit to the available data. Moreover, the lack of publicly available large datasets for construction safety monitoring as benchmarks causes difficulties when comparing the performance of various algorithms.

In addition to on-site cameras, numerous photos are taken daily on construction sites by workers and superintendents. Vision-based methods hold great potential for hazard identification and classification [15]. Future research may explore the possibility of crowdsourcing as a safety monitoring technique. Most photos taken by employees on site are unorganized and stored locally. Therefore, automatic hazard recognition, and frameworks for facilitating big data management can benefit construction industry in practice. Along this line, automatic image registration with BIM can also be explored as a future research direction.

Many construction companies today employ far-field cameras to capture images with a fixed time interval to create time-lapse videos, capturing the entire construction phase. While, these images can provide valuable information and insight for management, their application is currently limited to marketing and possibly conflict resolution [170]. Future research may investigate how the recent advancements in computer vision and machine learning can be utilised to take advantage of such resource for safety improvement. For instance, construction companies often carry out multiple projects simultaneously. Therefore, a comparative safety risk analysis of various construction projects based on real-time information can assist management in decision making regarding safety and resource allocation.

6. Conclusion

Developing sensor-based systems for improving construction safety has recently gained considerable momentum in the literature. In particular, vision-based monitoring systems have become mainstream research in construction. This study has provided an overview of recent advancements in the domain as they relate to safety risk management and has identified opportunities for future work.

Safety risk management as described by Safe Work Australia [10] includes four steps, namely identifying hazards, assessing risks, controlling risks, and reviewing control measures. The impact of sensor-based safety management systems on each aspect of the process, along with examples of various methods employed by researchers has been discussed. The reviewed studies cover a range of technologies being used. These include but are not limited to the utilisation of wireless sensor networks, RFID, UWB, IMUs (accelerometers and gyroscopes), EEG, and vision-based techniques for location tracking, gait analysis, object detection, activity recognition, ergonomic assessment, and so on for the development of sensor-based safety management systems.

Sensor-based systems can be used for automated continuous monitoring of construction sites. They can be used to proactively prevent accidents by notifying workers or equipment operators of an incoming hazard. Further, such systems can be used to notify safety managers or site supervisors of unidentified, or newly appearing hazards. As a result, necessary controls can be applied in a timely manner to prevent accidents.

Further investigations involving BIM revealed the potential to enable its integration with sensor-based systems. The traditional methods of safety management highly rely on the experience of the safety manager and workers, involve manual procedures, and are highly inefficient [61]. A dedicated safety BIM dimension, incorporating various sensor-based systems, can significantly improve the effectiveness of safety performance, by addressing the problems of traditional safety management such as time-consuming, labor-intensive and subjective aspects of the process.

Based on the reviewed literature, several gaps and future research directions were identified and discussed. Future research may investigate methods of safety risk quantification for automatic identification and documentation of near-miss incidents. Integration of BIM and sensor-based systems enable development of decision support platform that future research can explore. In addition, due to the abundance of visual data and the visual complexities that are unique to construction environments, computer vision research in the context of construction management is expected to increase.

Although a considerable amount of publications are reviewed to cover sensor-based systems thoroughly, future research should examine state-of-the-art techniques and technologies used in other sectors and consider their applicability to tackling construction safety challenges. The overview provided in this study can be used as a guide for researchers in pursuit of developing intelligent sensor-based safety management systems for improving safety performance in the construction industry.

Declaration of competing interest

All authors have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version.

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